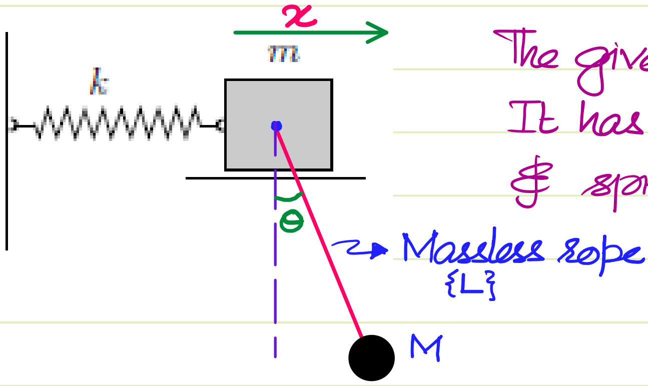
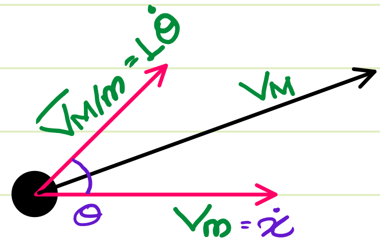


The given system has 2DOF $\{x, \theta\}$
It has a block $\{m\}$ & a particle $\{M\}$
& spring constant $\{k\}$.
Identify eq^{ns} of Motion.

Sol:

$U = U_{spring} + U_M$

$U = \frac{1}{2}kx^2 - MgL\cos\theta$ —— ①



$K = K_m + K_M$

$K_m = \frac{1}{2}m\dot x^2$

$K_M = \frac{1}{2}M\{\dot x^2 + L^2\dot\theta^2 + 2L\dot x\dot\theta\cos\theta\}$

$\vec V_M = \vec V_m + \vec V_{M/m}$

$V_M^2 = \dot x^2 + \{L\dot\theta\}^2 + 2\dot x(L\dot\theta)\cos\theta$

$= \dot x^2 + L^2\dot\theta^2 + 2L\dot x\dot\theta\cos\theta$

$$\boxed{L = \frac{1}{2}\{M+m\}\dot x^2 + \frac{1}{2}ML^2\dot\theta^2 + ML\dot x\dot\theta\cos\theta - \frac{1}{2}kx^2 + MgL\cos\theta}$$

Eq^{ns} of Motion:

$\frac{\partial L}{\partial \dot x} = (M+m)\dot x + ML\dot\theta\cos\theta$

$\frac{d}{dt}\left\{\frac{\partial L}{\partial \dot x}\right\} = \{M+m\}\ddot x + ML\ddot\theta\cos\theta - ML\dot\theta^2\sin\theta$

$\frac{\partial L}{\partial x} = -kx$

$\frac{d}{dt}\left\{\frac{\partial L}{\partial \dot x}\right\} - \frac{\partial L}{\partial x} = 0$

$\Rightarrow \boxed{\{M+m\}\ddot x + ML\ddot\theta\cos\theta - ML\dot\theta^2\sin\theta + kx = 0}$

$\frac{\partial L}{\partial \dot\theta} = ML^2\dot\theta + ML\dot x\cos\theta$

$\frac{d}{dt}\left\{\frac{\partial L}{\partial \dot\theta}\right\} = ML^2\ddot\theta + ML\ddot x\cos\theta - ML\dot x\dot\theta\sin\theta$

$\frac{\partial L}{\partial \theta} = -ML\dot x\dot\theta\sin\theta - MgL\sin\theta$

$\frac{d}{dt}\left\{\frac{\partial L}{\partial \dot\theta}\right\} - \frac{\partial L}{\partial \theta} = 0$

$\Rightarrow ML^2\ddot\theta + ML\ddot x\cos\theta + MgL\sin\theta = 0$

$\Rightarrow \boxed{L\ddot\theta + \ddot x\cos\theta + g\sin\theta = 0}$

4.4. Impulse Momentum Th^{rm}.

4.4.1. Linear Impulse & Momentum

Linear Impulse : The product of force & the duration over which the force acts. It is a vector quantity & has the direction of force.

$$\boxed{\vec{I}_F = \vec{F}.\Delta t}$$

S.I : N.s = kg m/s

Linear Momentum : The product of Mass & velocity of a body. It is a vector quantity with direction along Velocity Vector.

$$\boxed{\vec{P} = m\vec{V}}$$

S.I : kg m/s = N.s.

Impulse - Momentum Th^{rm} :

by Newtons' 2nd Law of Motion :

$$\vec{F} = m\vec{a} \Rightarrow \vec{F} = m\left\{\frac{d\vec{V}}{dt}\right\} \Rightarrow \boxed{\vec{F} = \frac{d\{m\vec{V}\}}{dt}}$$

⇒ The rate of change of Linear Momentum of a body is equal to the force acting on the body.

$$\boxed{\vec{F}.dt = d\{m\vec{V}\}}$$ Impulse - Momentum Eq^n.

Impulse on a body = Change in Linear Mom of the body

A particle moves in the x-y plane under the influence of a force such that its linear momentum is $\mathbf{p}(t) = A[\hat{\mathbf{i}}\cos(kt) - \hat{\mathbf{j}}\sin(kt)]$, where, A and k are constants. The angle between the force and the momentum is

(a) 0° (b) 30° (c) 45° (d) 90° ✓

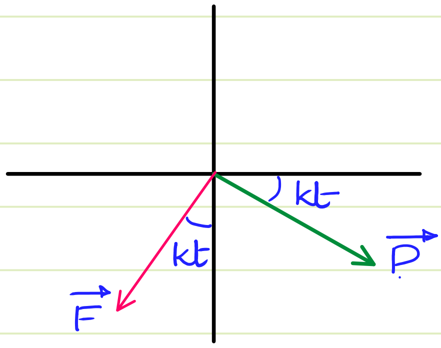
$$\vec{P} = A\{\cos(kt)\hat{i} - \sin(kt)\hat{j}\}$$

$$\vec{F} = \frac{d}{dt}(\vec{P})$$

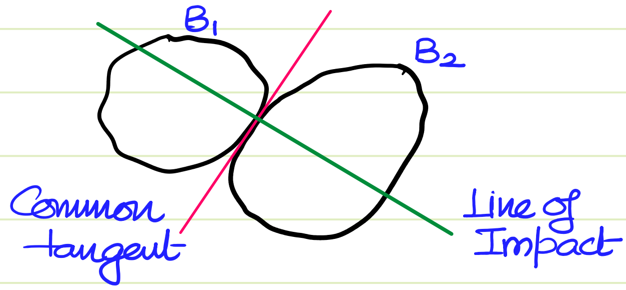
$$= Ak\{-\sin(kt)\hat{i} - \cos(kt)\hat{j}\}$$

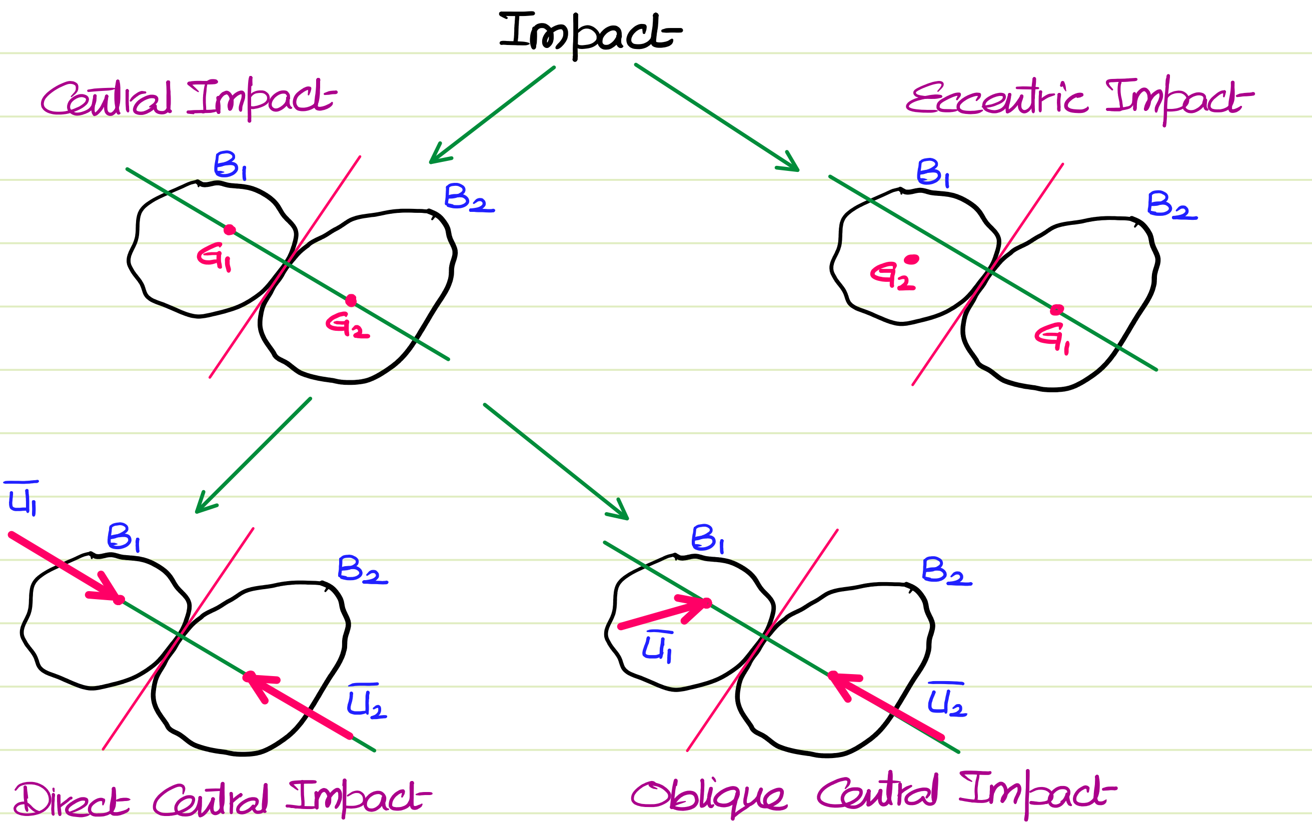
$\therefore$ $\vec{P}$ & $\vec{F}$ are orthogonal to each other.

If kt is an acute angle



Impact: Collision b/w 2 bodies.
The line $\perp^{ar}$ to the common tangent at the pt. of contact is known as Line of Impact.

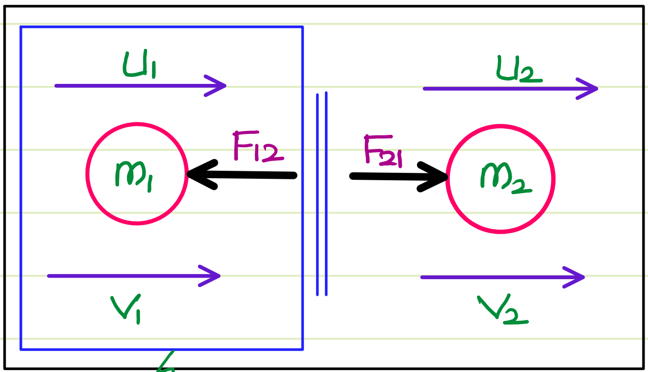




Law of Conservation of Linear Momentum :

The net linear Momentum of a system remains conserved if the net external force on the system is zero.

$\vec{F} = 0 \longrightarrow \vec{F}.\Delta t = 0 \longrightarrow \Delta(m\vec{v}) = 0 \longrightarrow m\vec{v} = c$



$\vec{F_{12}}$ & $\vec{F_{21}}$ are forming ac^n reac^n pair

$\Rightarrow$ for the net system $\Sigma F_{ext} = 0$

$\Rightarrow$ Momentum = Const.

$$\boxed{m_1\bar{u}_1 + m_2\bar{u}_2 = m_1\bar{v}_1 + m_2\bar{v}_2}$$

for this system $\{\Sigma F_{ext} = \vec{F_{12}}\} \neq 0$ $\therefore$ Momentum isn't consv.

By Impulse - Momentum th^rm : $\boxed{m_1\bar{v}_1 - m_1\bar{u}_1 = \bar{F}_{12}.\Delta t}$

Coefficient of Restitution : Every impact has 2 stages

1. Deformation of body.
2. Restitution / Restoration of body.

Timeline of impact :

$t=0$ ——— $t=t'$ ——— $t=T$

Deformation {D} (between $t=0$ and $t=t'$); Restitution {R} (between $t=t'$ and $t=T$)

The ratio of Impulse of Restitution $\left\{\int_{t'}^{T} R.dt\right\}$ to the Impulse of deformation $\left\{\int_{0}^{t'} D.dt\right\}$ is known as Coeff. of Restitution.

$$e = \frac{\int_{t'}^{T} R.dt}{\int_{0}^{t'} D.dt} = \frac{\text{Relative Velocity after Impact}}{\text{Relative Velocity before Impact}} \Rightarrow \boxed{e = \frac{|\bar{v}_2 - \bar{v}_1|}{|\bar{u}_2 - \bar{u}_1|}}$$

Classification of Impact based on Coeff. of Restitution :

1. Plastic Impact $\{e=0\}$ Both bodies move together after impact.
   * ∵ there is no restoration, ∴ Max deformation, ∴ Max strain energy. ⇒ Max loss in K.E.
   * Linear Momentum of the System can be conserved

2. Inelastic Impact : $\{0<e<1\}$ Partial restoration of the bodies.
   * Certain amount of kinetic energy is lost.
   * Linear Momentum of the System can be conserved

3. Elastic Impact : $\{e=1\}$ Complete restoration of the bodies.
   * K.E. of the system can be conserved
   * Linear Momentum of the System can be conserved

A particle of mass m moving in the x-direction with speed $2v$ is hit by another particle of mass $2m$ moving in the y-direction with speed v. If the collision is perfectly inelastic, the percentage loss in the energy during the collision is close to

(a) 50 % (b) 56 % (c) 62 % (d) 44 %

By Law of Cons. of P in x :

$m\{2v\} + 2m(0) = 3mV_x$

$\Rightarrow V_x = 2v/3$

By Law of Cons. of P in y :

$m\{0\} + 2m(v) = 3mV_y$

$\Rightarrow V_y = 2v/3$

$K_{initial} = \frac{1}{2}m(2v)^2 + \frac{1}{2}2mv^2$

$= 3mv^2$

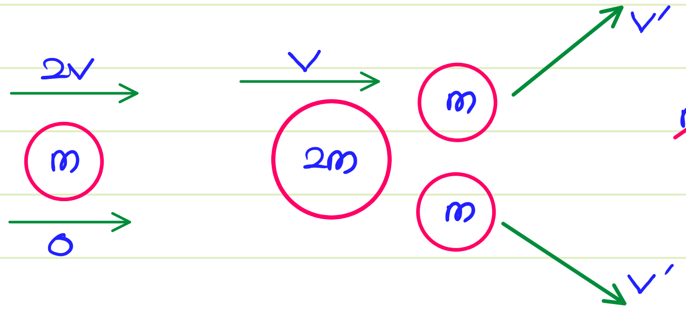
$K_{final} = \frac{1}{2}\{3m\}\{V_x^2 + V_y^2\} = \frac{3m}{2}\left\{\frac{8v^2}{9}\right\}$

$= \frac{4mv^2}{3} \Rightarrow K_{loss} = \frac{5mv^2}{3}$

$\% \text{ loss} = \frac{\frac{5mv^2}{3}}{3mv^2} \times 100\% \Rightarrow \boxed{\% \text{ loss} = 55.55\%}$

A particle of mass m is moving with speed $2v$ and collides with a mass 2m moving with speed v in the same direction. After collision, the first mass is stopped completely while the second one splits into two particles each of mass m, which move at angle 45° with respect to the original direction. The speed of each of the moving particle will be

(a) $\sqrt{2}v$ (b) $\frac{v}{\sqrt{2}}$ (c) $\frac{v}{2\sqrt{2}}$ (d) $2\sqrt{2}v$



By consv. Momentum in x

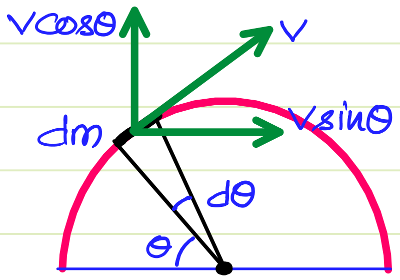
$\cancel{m}(2V) + 2\cancel{m}(V) = 2\cancel{m}V'\cos 45°$

$2V = V'/\sqrt{2}$

$\Rightarrow V' = 2\sqrt{2}\,V.$

A train of mass **M** is moving on a circular track of radius **R** with constant speed **v**. The length of the train is half of the perimeter of the track. The linear momentum of the train will be

(a) 0
(b) $\left\{\frac{2Mv}{\pi}\right\}$
(c) Mv
(d) πMv



$\frac{dm}{R.d\theta} = \frac{M}{\pi R}$

$\Rightarrow dm = \frac{M}{\pi}.d\theta$

$d\vec{P} = \frac{M}{\pi}.d\theta\{V\sin\theta\,\hat{i} + V\cos\theta\,\hat{j}\}$

$\vec{P} = \frac{MV}{\pi}\left\{\int_0^\pi \sin\theta.d\theta\,\hat{i} + \int_0^\pi \cos\theta.d\theta\,\hat{j}\right\} \Rightarrow \vec{P} = \frac{2MV}{\pi}\hat{i}$

A man (mass = 50 kg) and his son (mass = 20 kg) are standing on a frictionless surface facing each other. The man pushes his son, so that he starts moving at a speed of 0.70 ms^{-1} with respect to the man. The speed of the man with respect to the surface is

(a) 0.28 ms^{-1} (b) 0.20 ms^{-1} (c) 0.47 ms^{-1} (d) 0.14 ms^{-1}

Initial Mom. of system $= 0$

Final Mom $= 50V_M + 20V_S$

$\Rightarrow 0 = 50V_M + 20V_S$

$\Rightarrow V_M = -\frac{2}{5}V_S.$

Given: $V_{S/M} = 0.7\,m/s \Rightarrow V_S - \{-\frac{2}{5}V_S\} = 0.7\,m/s \Rightarrow V_S = 0.5\,m/s$

$\therefore V_M = -0.2\,m/s.$

A ball of mass 3 kg moving with a velocity of 4 m/s undergoes a perfectly-elastic direct-central impact with a stationary ball of mass m. After the impact is over, the kinetic energy of the 3 kg ball is 6 J. The possible value(s) of m is/are

(A) 1 kg
(B) 3 kg
(C) 6 kg
(D) 9 kg

The Velocity of 3kg ball after impact : $\frac{1}{2}\{3\}V^2 = 6$

$\Rightarrow V^2 = 4$

$\Rightarrow \boxed{V = 2m/s.}$

Initial K.E. of the system $= \frac{1}{2} \times 3 \times 4^2 = 24J.$

1st Possibility : 3kg ball Maintains dir^n sense.

$3(4) + m(0) = 3(2) + mv$

$\Rightarrow mv = 6$

KE of m after Impact $= \frac{1}{2}mv^2 = 18J$

$\Rightarrow \frac{1}{2}mv.v = 18$

$v = 18 \times 2/6$

$\Rightarrow v = 6 \quad \therefore m = 1kg$

2nd possibility : The dir^n of 3 kg ball is reversed.

$3(4) + m(0) = 3(-2) + mv$

$\Rightarrow mv = 18$

KE of m after Impact $= \frac{1}{2}mv.v = 18$

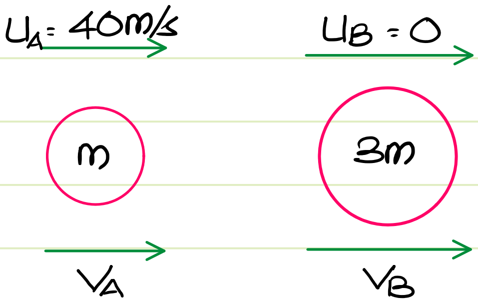
$\Rightarrow v = 2m/s$

$\therefore m = 9kg$

A sphere **A** of mass m is thrown into the air at **50 m/s** along a direction **$\tan^{-1}$ (3/4)** up from the horizontal. At the topmost point of its trajectory, it has a central (non-oblique) collision with another sphere **B** which is at rest on top of a vertical pole. Sphere B has a mass of **3m**. Acceleration due to gravity is **10 m/s²**. Neglect contact friction and air-resistance. If the coefficient of restitution is **0.3**, then the speed (in m/s) of sphere A immediately after the collision is ……… *(rounded off to one decimal place)*

$\theta = \tan^{-1}\{3/4\} \Rightarrow \cos\theta = 4/5$

Vel. of A @ top of traj $= 50\cos\theta$
$= 40$ m/s



$e = \dfrac{|\bar{V}_B - \bar{V}_A|}{|\bar{U}_B - \bar{U}_A|}$

$\Rightarrow V_B - V_A = 0.3 \times 40$

$\therefore V_B - V_A = 12$ m/s

$V_B = V_A + 12$

By Law of Consv. of Mom:

$m(40) + 3m(0) = mV_A + 3mV_B$

$\Rightarrow 40 = V_A + 3\{V_A + 12\}$

$\Rightarrow 4V_A = 4$

$\therefore V_A = 1$ m/s

Note: When problems involving Impact followed by Motion are given, break down the solution into 2 parts.

1. during Impact apply Impulse Momentum th^{rm} (or) Law of Consv. of Momentum.
2. during Motion prefer Work - Energy th^{rm}/Law of Conservation of Energy / Newton's 2nd Law of Motion.

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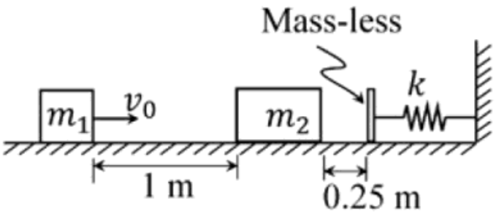
A rigid block of mass $m_1 = 10$ kg having velocity $v_0 = 2$ m/s strikes a stationary block of mass $m_2 = 30$ kg after traveling 1 m along a frictionless horizontal surface as shown in the figure.



The two masses stick together and jointly move by a distance of 0.25 m further along the same frictionless surface, before they touch the mass-less buffer that is connected to the rigid vertical wall by means of a linear spring having a spring constant $k = 10^5$ N/m. The maximum deflection of the spring is ________________ cm (*round off to 2 decimal places*).

Law of Conservation of P during Impact :

$$m_1 V_0 + m_2(0) = (m_1 + m_2)V$$

$$\Rightarrow V = \frac{10 \times 2}{40}$$

$$\therefore V = 0.5 \, m/s$$

For Max compression of spring

$$K_{M_1+M_2} = U_{spring}$$

$$\Rightarrow \frac{1}{2}(M_1+M_2)V^2 = \frac{1}{2}k x^2$$

$$\Rightarrow 10 = 10^5 . x^2$$

$$\therefore x^2 = 10^{-4} m^2$$

$$\therefore x = 10^{-2} m = 1cm$$

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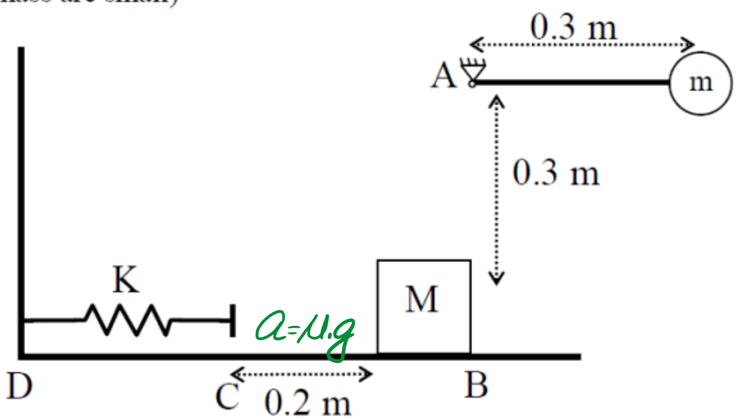
A striker with mass m=20 kg is attached to the end of a mass-less rigid bar of length R=0.3m. The bar is hinged to support A, and swings down from an initial horizontal position such that the striker hits mass M= 5kg elastically. The mass M slides (in a straight line) along the table, from point B towards the spring located at point C, 0.2 m away from B. Assume that the coefficient of friction in the region BC is $\mu_d = 0.4$; the region CD frictionless. Let the spring constant of the spring be k = 4000 N/m. If the striker rises to a maximum height of 0.1m below its starting location, then the maximum compression of the spring is (let acceleration due to gravity g = 10 m/s^2, the dimensions of the striker and mass are small)



(A) 0.000 m (B) 0.100 m (C) 0.089 m (D) 0.109 m

∵ the collision b/w m & M is elastic, ∴ Energy can be cons.

$$mg\{0.3\} = mg\{0.2\} + \frac{1}{2}M.V^2$$

$$\therefore V = \sqrt{\frac{2mg(0.1)}{M}} = 2\sqrt{2}\ m/s$$

Motion from B to C : $V_C^2 - V_B^2 = 2a.s$

$$\Rightarrow V_C^2 = V_B^2 + 2as$$

$$= (2\sqrt{2})^2 + 2(-0.4\times 10)(0.2)$$

$$V_C^2 = 8 - 1.6 \qquad \therefore V_C = 8/\sqrt{10}$$

for Max comp. $K_M = U_{spring}$

$$\cancel{\tfrac{1}{2}}M.V_C^2 = \cancel{\tfrac{1}{2}}kx^2 \Rightarrow 5\times 6.4 = 4000.x^2$$

$$\therefore x = 0.08944\,m$$

4.4.2. Angular Impulse & Momentum

Angular Impulse {or} Impulse of Moment $\{\vec{I}_M\}$

The product of Moment or Torque & the duration of its action

$$\boxed{\vec{I}_M = \vec{M}.\Delta t}$$ S.I : N.m.s = kg m²/s

The rotational effect of Impulse of force also results in Angular Impulse

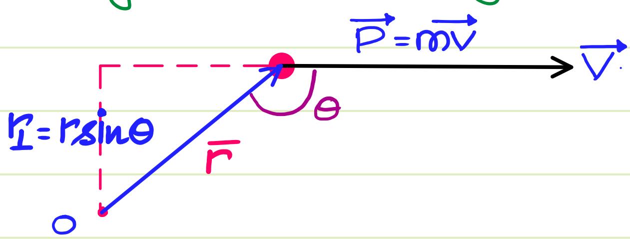
$$\boxed{\vec{I}_M = \vec{r} \times \vec{I}_F}$$ S.I. : N.m.s = kg m²/s

Angular Momentum {or} Moment of Momentum $\{\vec{H}\}$

The moment of linear Momentum of a body about a point {or} axis is Angular Momentum. Therefore Angular Momentum is the rotational effect of Linear Momentum.

$$\boxed{\vec{H} = \vec{r} \times \vec{P} = \vec{r} \times m\vec{v}}$$ S.I. : N.m.s = kg m²/s

1: Angular Momentum of a body in Pure Translation :



$$H_O = \vec{r} \times m\vec{v} = mv\boxed{r\sin\theta} \to r_\perp$$

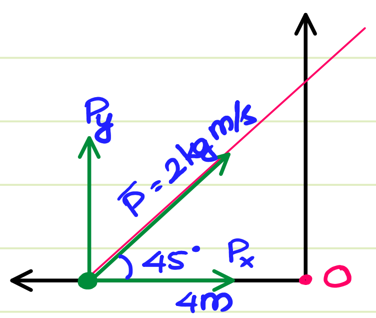
$$\therefore \boxed{H_O = mvr_\perp}$$

IM A particle of mass 1 kg is moving along a straight line $y = x + 4$. Both x and y are in metres. Velocity of the particle is 2 m/s. Find magnitude of angular momentum of the particle about origin.

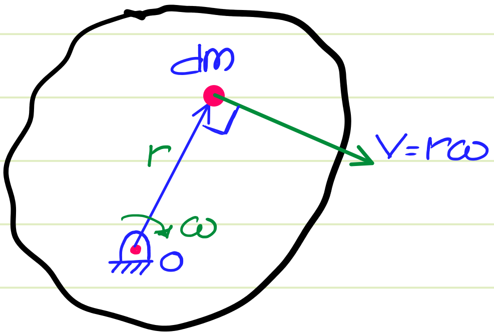
$$H_O = \cancel{H_O^{P_x}} + H_O^{P_y}$$

$$= 2\sin 45^\circ \times 4m$$

$$= 4\sqrt{2} \text{ kg m}^2/\text{s}$$



2. Angular Momentum of A Body In Pure Rotation {FAR}



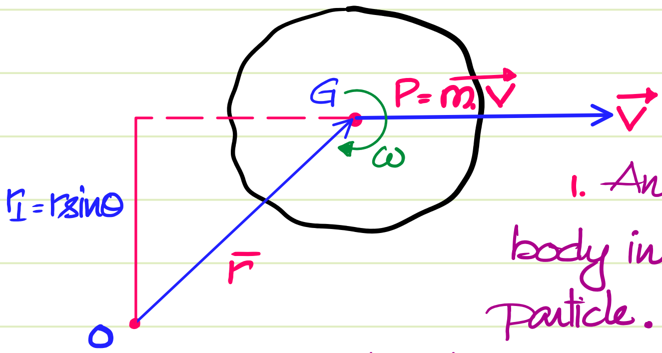
$dH_O = dm\,v.r$

$= dm.r\omega.r$

$\Rightarrow H_O = \omega \int dm.r^2$

$$\boxed{\vec{H}_O = I_O.\omega}$$

3. Angular Momentum of A body in GPM {Translation + Rotation}



Breakdown the calculation into 2 parts :

1. Angular Momentum of the entire body in pure translation as if it is a particle. $H_T = mvr_\perp$
2. Angular Momentum of the body due to rotation abt centre of Gravity $H_R = I_G.\omega$

$$\boxed{\vec{H}_O = m\vec{v}r_\perp + \vec{I_G.\omega}}$$

Note : During GPM, the angular Momentum about any random point can't be taken as $I\omega$

$\cancel{H_O = I_O.\omega}$

But this formula is Valid if that point is I Centre

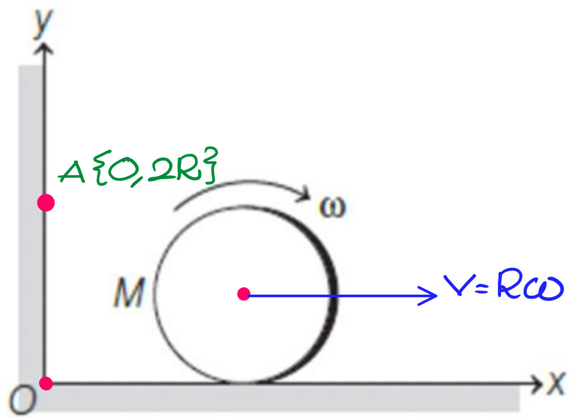
$\Rightarrow \bar{H}_I = I_I.\omega$

$$\boxed{\bar{H}_I = m\bar{v}r_\perp + \overline{I_G.\omega} = I_I.\omega}$$

A disc of mass M and radius R is rolling with angular speed ω on a horizontal plane as shown. The magnitude of angular momentum of the disc about the origin O is

(a) $\frac{1}{2}MR^2\omega$ (b) $MR^2\omega$

(c) $\frac{3}{2}MR^2\omega$ (d) $2MR^2\omega$



$$\vec{H}_O = \vec{H}_T + \vec{H}_R$$
$$= m\vec{v}\vec{r}_\perp + I_G.\vec{\omega}$$
$$= MR\omega R + \frac{MR^2}{2}.\omega$$

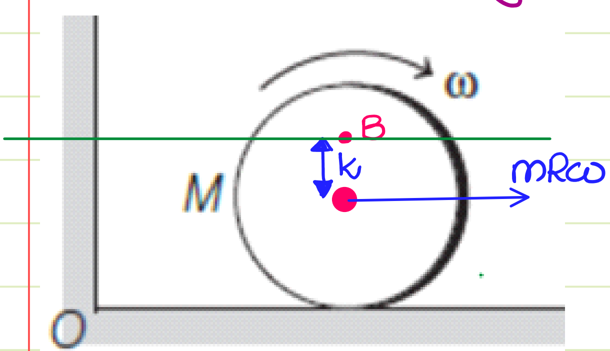
$$\vec{H}_O = \frac{3}{2}MR^2\omega$$

Find the value of Angular Momentum about pt. A.

$$\vec{H}_A = \vec{H}_T + \vec{H}_r$$
$$= m\vec{v}\vec{r}_\perp + I_G.\vec{\omega}$$
$$= MR\omega R + \frac{MR^2}{2}.\omega$$

$$\vec{H}_A = \frac{1}{2}MR^2\omega$$

In the above case, find the eq^n of line such that angular Momentum about any point on that line is zero.



$$H_B = m\vec{v}r_\perp + I_G.\omega = 0$$
$$MR\omega.r_\perp + \frac{MR^2\omega}{2} = 0$$
$$r_\perp = R/2$$

Eq^n of line : $y = 3R/2$

Note : Any amount of force applied along this line gives rolling w/o slip with zero friction.

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Physical Significance of Angular Momentum :

$\vec{H} = \vec{r} \times m\vec{v}$

By differentiation wrt time : $\frac{d\bar{H}}{dt} = \frac{d}{dt}\{\bar{r} \times m\bar{v}\}$

$\Rightarrow \frac{d\bar{H}}{dt} = \left\{\bar{r} \times \frac{d\{m\bar{v}\}}{dt}\right\} + \left\{\frac{d\bar{r}}{dt} \times m\bar{v}\right\}$

$= \{\bar{r} \times \bar{F}\} + \{\bar{v} \times m\bar{v}\}$ $\{\because \bar{v}$ & $m\bar{v}$ are like vectors $\therefore \bar{v} \times m\bar{v} = 0\}$

$\Rightarrow \boxed{\frac{d\bar{H}_A}{dt} = M_A}$ $\Rightarrow$ The rate of change of angular Momentum about a point is Torque abt that point.

Angular Impulse - Angular Momentum Th^ron :

$\boxed{\vec{M}_A . \Delta t = \Delta\vec{H}_A}$ $\Rightarrow$ Angular Impulse - Angular Mom. Th^ron.

$\Rightarrow$ Impulse of Moment = Change in Angular Momentum.

Law of Conservation of Angular Momentum :

When the net torque on a system is zero about a particular point, then the angular momentum about that point remains conserved.

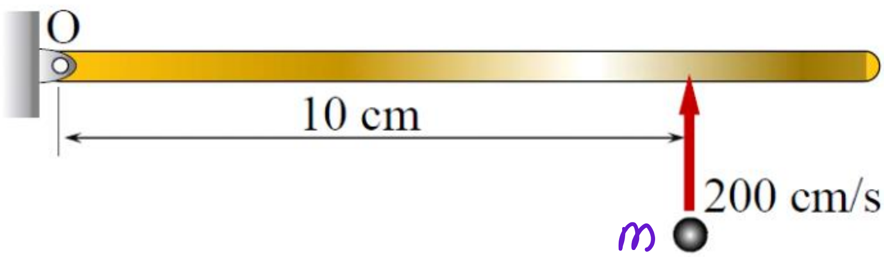
$$\bar{M}_A = 0 \Rightarrow \bar{H}_A = \text{constant}$$

Case Study : If ΣF on a system $\neq 0 \Rightarrow$ Linear Mom. isn't conserv.

If a point or axis is chosen such that $\Sigma M = 0 \Rightarrow$ Angular Mom. abt. that pt. is Cons.

Note : Angular Momentum can be conserved despite Linear Momentum not being conserved.

IM. The plane of the figure represents a horizontal plane. A thin rigid rod at rest is pivoted without friction about a fixed vertical axis passing through O. Its mass moment of inertia is equal to 0.1 kg·cm^2 about O. A point mass of 0.001 kg hits it normally at 200 cm/s at the location shown, and sticks to it. Immediately after the impact, the angular velocity of the rod is ___________ rad/s (*in integer*).



KE can't be cons. ∵ plastic Impact

$\bar{P}$ can't be cons ∵ Hinge reac^n are present.

∵ ΣF passes thru O ∴ $\Sigma M_O = 0 \Rightarrow \bar{H}_O$ can be cons.

$$\{\bar{H}_O\}_{initial} = \{\bar{H}_O\}_{final}$$
$$m v r_\perp = \{I_o . \omega\}_{rod} + m R^2 \omega$$
$$10^{-3} \times 200 \times 10 = 0.1\omega + 10^{-3} \times 10^2 \omega$$
$$\Rightarrow 2 = 0.2\omega$$

$\boxed{\omega = 10 \text{ rad/s}}$

Central force system: If a body experiences only one force about a fixed point $\{\Sigma F \neq 0\}$, then Linear Mom can't be conserved, but the net Moment about that point must be zero. $\{\Sigma \bar{M} = 0\}$ The angular Momentum about that point is conserved.

IM. The angular momentum of a particle moving under a central force is

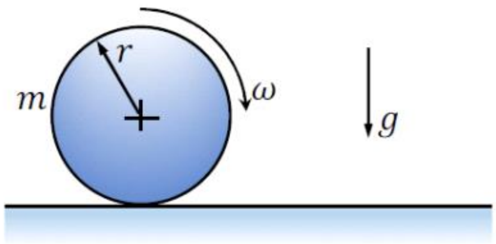
(A) zero.

(B) constant in both magnitude and direction. ✓

(C) constant in magnitude but not direction.

(D) constant in direction but not magnitude.

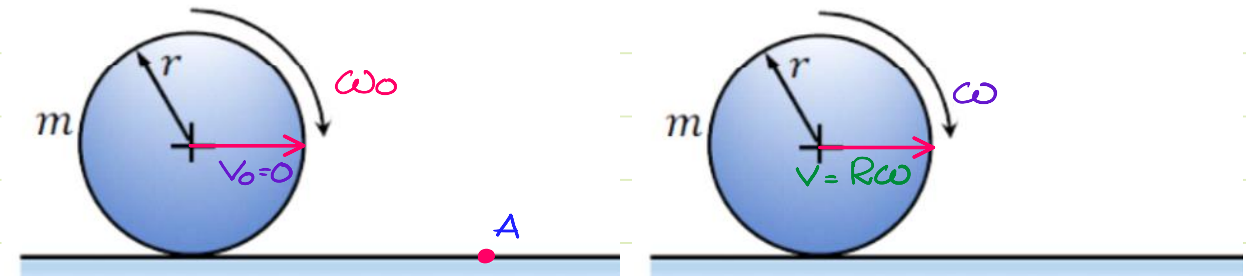
A cylindrical disc of mass $m = 1$ kg and radius $r = 0.15$ m was spinning at $\omega = 5$ rad/s when it was placed on a flat horizontal surface and released (refer to the figure). Gravity g acts vertically downwards as shown in the figure. The coefficient of friction between the disc and the surface is finite and positive. Disregarding any other dissipation except that due to friction between the disc and the surface, the horizontal velocity of the center of the disc, when it starts rolling without slipping, will be __________ m/s (*round off to 2 decimal places*).



Is KE conserved : No $\{W_{friction} = -ve \ \therefore \text{KE loss}\}$

Is Linear Mom Conserved : No $\{\Sigma F = f\} \neq 0$

$\because \Sigma F = f$, $\therefore$ The Net Moment about any point along the line of friction is zero $\therefore$ Angular Mom about any point along that line can be conserved.



By Cons. of angular Mom abt A :

$$\{H_A\}_{initial} = \{H_A\}_{final}$$

$$I_G \cdot \omega_0 = I_G \cdot \omega + mVR$$

$$\frac{MR^2}{2} \cdot \omega_0 = \frac{MR^2}{2}\omega + MR^2\omega \quad \Rightarrow \omega = \omega_0/3$$

$$\therefore V = \frac{R\omega_0}{3}$$

$$\therefore V = 0.25 \text{ m/s.}$$

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Find the work done by friction ?

W_f = Loss in KE.

$= KE_i - KE_f$

$= \frac{1}{2} I_G . \omega_0^2 - \frac{1}{2} I_I . \omega^2$

$= \frac{1}{2} \times \frac{MR^2}{2} . \omega_0^2 - \frac{1}{2} \left\{ \frac{3MR^2}{2} \right\} \left\{ \frac{\omega_0}{3} \right\}^2$

$= \frac{MR^2 \omega_0^2}{4} - \frac{MR^2 \omega_0^2}{12}$

$= \frac{MR^2 \omega_0^2}{6} = \frac{1 \times 0.15^2 \times 5^2}{6}$

$= 0.09375 J.$